

Question	Marking Guidance	Mark	Comments
10.1	1. No (functional) cones OR Only rods; 2. Cones are connected to a single neurone OR Several rods connected to a single neurone; 3. (Cones) Separate (sets of) impulses to brain OR (Rods) Single (set of) impulse/s to brain;	3	2. Accept correct reference to retinal convergence 2. Accept 'bipolar/nerve cell' for neurone 2. Accept 'many' 2 or more for 'several' 3. Accept 'optic nerve' for brain 3. Reject 'signals', 'messages' for 'impulses' 3. Accept 'action potential'
10.2	1. Correct answer in range 42 – 44% = 2 marks;; 2. Incorrect answer but shows that understanding that $2pq$ = heterozygous/carriers = 1 mark;	2	2. Accept $1 - (p^2 + q^2)$ 2. Accept understanding of $2pq$ by using calculation involving $2 \times$ two different numbers
10.3	1. (Gene/allele) is on the <u>X chromosome</u>; 2. Females require two alleles/females can be heterozygous/carriers and males require one allele;	2	2. Reference to allele is essential but only required once 2. Reference to females and males required 2. Reject dominant allele
10.4	1. Green sensitive pigment/cones non-functional OR Cones that detect green light non-functional; 2. Three different types of pigment/cone;	3	1, 2 and 3. Reject reference to 'green cones' / 'blue cones' / 'red cones' but once only 1, 2 and 3. Reject reference to 'green pigment' / 'blue pigment' / 'red pigment'

	3. Other/different colours ('seen') due to stimulation of more than one cone/pigment;		but once only and only if 'green cones' etc, (see above) has not been rejected.
10.5	1. (iPS cells) divide; 2. (iPS cells) develop/differentiate into (green sensitive) cones;	2	2. Accept 'produce'/'specialise' 'turn in to' / 'genes switched on' / 'turned on' for 'develop' but ignore 'grow' 2. Reject develop into 'green cones'/blue' cones'/'red cones' Ignore develop/differentiate into (blue/red sensitive) cones; Reject reference to develop in to 'green pigment'/'blue pigment'/'red pigment'

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10.6	1 (Use of iPS cells) long-term; 2. (Use of iPS cells) less chance of rejection/immune response; 3. (Use of iPS cells) single treatment; 4. Harm/side effects from using viruses (in gene therapy);	3 max	1. Accept 'gene therapy short-term' or 'only two years' 1. Accept 'permanent' 3. Accept 'gene therapy 'regular/frequent treatment''